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The multiscale entropy: Guidelines for use and interpretation in brain signal analysis

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HIGHLIGHTS

- We provide an intuitive explanation of MSE, its application and interpretation.
- Both simulated and empirical data are used.
- We show how the properties of a signal are reflected in the MSE curves.
- MSE captures linear and nonlinear autocorrelations.
- The use of complementary methods helps in preventing misleading interpretations.

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ABSTRACT

Background: Multiscale entropy (MSE) estimates the predictability of a signal over multiple temporal scales. It has been recently applied to study brain signal variability, notably during aging. The grounds of its application and interpretation remain unclear and subject to debate.
Method: We used both simulated and experimental data to provide an intuitive explanation of MSE and to explore how it relates to the frequency content of the signal, depending on the amount of (non)linearity and stochasticity in the underlying dynamics.
Results: The scaling and peak-structure of MSE curves relate to the scaling and peaks of the power spectrum in the presence of linear autocorrelations. MSE also captures nonlinear autocorrelations and their interactions with stochastic dynamical components. The previously reported crossing of young and old adults' MSE curves for EEG data appears to be mainly due to linear stochastic processes, and relates to young adults' EEG dynamics exhibiting a slower time constant.
Comparison with existing methods: We make the relationship between MSE curve and power spectrum as well as with a linear autocorrelation measure, namely multiscale root-mean-square-successive-difference, more explicit. MSE allows gaining insight into the time-structure of brain activity fluctuations. Its combined use with other metrics could prevent any misleading interpretations with regard to underlying stochastic processes.
Conclusions: Although not straightforward, when applied to brain signals, the features of MSE curves can be linked to their power content and provide information about both linear and nonlinear autocorrelations that are present therein.

1. Introduction

Multiscale entropy (MSE) has been used lately as a complexity measure for neuroimaging signals (for examples see Escudero et al., 2006; McIntosh et al., 2008, 2014; Sleimen-Malkoun et al., 2015; Yang et al., 2013). The purpose behind its use is to evaluate

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the structure of variability of such signals across multiple temporal scales. Costa et al. (2002, 2005), who introduced first this method and used it with biological signals, relate the degree of complexity of a time series to the existence of long-range temporal correlations. In brain signals, assessing the presence of temporal relationships over different scales is of interest as it provides important information about network dynamics (Heisz et al., 2012; Lopes da Silva, 2013; Stam, 2005). In this context, MSE is thought to inform about the relative contribution of local and global (long-range) information processing in the brain (McIntosh et al., 2014; Mizuno et al., 2010; Vakorin et al., 2011). Specifically, in M/EEG studies, the structure of variability at short time scales, or high frequencies, has been linked to local neural population processing, whereas variability at longer time scales, or lower frequencies, has been linked to large-scale network processing (McIntosh et al., 2014; Mizuno et al., 2010; Vakorin et al., 2011). This view was supported by the presence of (anti-)correlations between the changes of MSE values at short and long time scales with various measures of local and distributed entropy or functional connectivity, respectively (McDonough and Nashiro, 2014; McIntosh et al., 2014; Vakorin et al., 2011). This link between high/low frequencies and spatial scales of neuronal processing could be considered as more of a neurophysical than neurophysiological nature. Indeed, frequencies in different bands and across bands are associated with communication and information processing across spatial scales within the brain network. Synchronization and desynchronization, as well as its transient and multi-frequency organization capture the network's information flow and can be measured via cross-frequency coupling (CFC) (Jirsa and Müller, 2013). Although the existence of different types of CFC (such as phase-to-power or power-to-frequency) renders communication analysis highly nontrivial, the association of long-range communication with lower frequencies (delta, theta and alpha) and short-range communication with higher frequencies (beta and gamma) can be intuitively understood on the basis of error propagation, in which the absolute value of variance has more detrimental effects upon faster oscillations. Furthermore, despite the fact that MSE appears to have promising applications to distinguish amongst populations, as young and older groups, and task conditions, as rest and task conditions, (see Sleimen-Malkoun et al., 2015 for an overview and an example), it is not evident how to interpret the features of MSE profiles beyond concluding that one signal is more or less "entropic" for shorter or longer time scales than the other. It is also neither evident how to explicitly relate MSE profiles with underlying brain dynamics and neurophysiological processes. Indeed, given the widely accepted functional significance of the dynamic formation and dissolution of local and global brain networks for cognitive processes (Schnitzler and Gross, 2005; Sporns, 2011; Varela et al., 2001), it is relevant to shed some light on the relationship between MSE's time scales and the frequency content of brain signals, as well as the presence of (non)linear correlations therein. Clarifying these issues would render the use and interpretation of MSE more intuitive and provide a more precise picture of the network extent of the underlying processes. In current literature, only very few studies explicitly explored these aspects. For instance, Bruce et al. (2009) reported a correlation between the single scale entropy metric and the power spectrum (PS) namely a negative correlation with delta power and a positive one with beta. The authors even concluded that changes in single scale entropy reflect changes in spectral content rather than changes in regularity. Later studies using MSE attempted to relate entropy at different time scales to changes in the frequency content of the underlying signal, suggesting that fast and slow frequencies relative power were associated with MSE values for fine and coarser time scales, respectively (McIntosh et al., 2014; Mizuno et al., 2010; Vakorin and McIntosh, 2012; Vakorin et al., 2011). On the other hand, the issue of MSE being a measure of nonlinear changes in the brain has been less investigated. For example, Park et al. (2007) have presented MSE as a means to evaluate the nonlinear structure of brain dynamics without, however, establishing any empirical link in that regard.

Technically, MSE is an extension of the sample entropy (SampEn) algorithm (Richman and Moorman, 2000) using a temporal coarse-graining procedure (Costa et al., 2002, 2005). It aims at evaluating the variability of biological signals across a range of temporal scales. SampEn as a measure has been developed to quantify the structure of variability of a time series in terms of its regularity (or inversely, irregularity) or predictability over time through the identification of reproducible patterns. Higher values of SampEn being indicative of lesser predictability, or what it is commonly referred to as more complexity. The added temporal coarse-graining procedure in MSE consists of downsampling successively the original signal by applying a moving average filter with nonoverlapping windows of increasing length for increasing time scales (see Section 2); in the frequency domain this acts as a low-pass filter (Govindan et al., 2007; Kaffashi et al., 2008; Valencia et al., 2009). On the one hand, such a coarse-graining procedure alleviates the signal from linear effects between consecutive samples, like those associated with observational noise at short time scales, thereby MSE can evaluate the remaining correlations of possibly nonlinear deterministic nature (Govindan et al., 2007; Kaffashi et al., 2008; Vakorin et al., 2011; Valencia et al., 2009). On the other hand, as the coarse-graining modifies the frequency content of the signal, it renders the relationship between MSE and other multiscale variability measures, such as PS, less evident. To help gain insight into this latter issue, and to guide the application and interpretation of MSE on brain signals, in the present study we systematically investigate how the scaling and peak structure of PS as well as the (non)linearity[1] of a time series are reflected in the MSE curve, in comparison with the other measures. This is achieved by analyzing numerically simulated phenomenological examples of stochastic and oscillatory processes with different levels of nonlinearity at given frequency spectra, in addition to experimental EEG data. In order to demonstrate the ability of MSE to evaluate nonlinear autocorrelations in addition to linear ones, we compare it to a multiscale measure of linear autocorrelations that we introduce, which we refer to as the multiscale root-mean-square-successive-difference (MRMSSD). This method applies the root-mean-square-successive-difference (RMSSD) algorithm over a range of scales obtained through coarse-graining. The "single scale" RMSSD has been widely used to assess temporal variability of heart-rate (Berntson et al., 2005; Owen and Steptoe, 2003), and more recently in neuroimaging data (Samanez-Larkin et al., 2010). In the light of the obtained results, we discuss the implications of our systematic investigations on functional processing in the brain.

The remainder of this paper is organized as follows. Section 2 provides an overview of our approach as well as of the SampEn and MSE algorithms. In Section 3, our results demonstrate the close relationship between MSE, PS and MRMSSD. We investigate the scaling properties of the tested signals and provide rules of thumb for relating the peak (and possibly the slope) of MSE curves with the frequency content of a signal. Finally, in Section 4, we discuss to what degree there is complementarity or redundancy of information between MSE and the PS, as well as the implications of our results for linking SampEn values at short and long time

[1] By nonlinearity in time series, we refer to the existence of nonlinear autocorrelations, i.e., to the existence of relationships in the signal with itself of a nonlinear functional form at some time scales. The process generating this kind of signal could be described by a Langevin equation (Langevin, 1908), and such nonlinearities result from nonlinear terms (i.e., of higher than first order), either in the deterministic or in the stochastic functions.

scales with local processing and large-scale network integration in the brain, respectively. It is noteworthy to mention that, although our methodological exploration is undertaken in the context of the application of MSE to neuroimaging data, the provided general guidelines can be extended to any continuous time series.

2. Materials and methods

2.1. Data

In order to investigate the properties of the MSE curves across temporal scales, such as peak structure and scaling, we studied MSE in comparison with PS and MRMSSD analysis on two types of data: simulated and experimental time series. The simulated data were generated by manipulating the frequency content, the linear autocorrelations, as well as the different degrees of nonlinearity and stochasticity of the generating dynamical processes.

2.1.1. Simulated data

2.1.1.1. Signals of linear stochastic processes. We generated different profiles of power-law (or colored) noise signals, i.e., signals of independent random samples and containing only linear autocorrelations. Time series comprising white, pink, brown, and blue noise were created using a Fourier filtering technique (Makse et al., 1996). This method consists of the following steps:

1. First, we generated a stationary uncorrelated random sequence $\eta(t)$ of normal distribution (mean $<\eta(t)>=0$ and standard deviation $\sigma_\eta = 1$), and we calculated its Fourier transform $\hat{\eta}(f)$.
2. The desired correlated signal $x(t)$ must exhibit correlations, which are characterized by a spectral power of a power-law form

$$S(f) = |\hat{x}(f)|^2 \sim f^{-\beta}, \tag{1}$$

where f is the frequency, $\hat{x}(f)$ is the Fourier transform of $x(t)$, and β is the spectral exponent.
Thus, we generated $\hat{x}(f)$ using the following transformation

$$\hat{x}(f) = S(f)^{1/2}\hat{\eta}(f) \sim f^{-\beta/2}\hat{\eta}(f), \tag{2}$$

where $S(f)$ is the desired spectrum in Eq. (1).
3. Finally, we computed the inverse Fourier transform of $\hat{x}(f)$ to obtain the time series $x(t)$.

The signal $x(t)$ so generated is a stationary power-law correlated signal. Since the spectral power and the autocorrelation function of a (wide-sense-)stationary process form a Fourier transform pair, a relation known as the Wiener–Khinchin theorem (Wiener, 1930), the amplitude of the autocorrelation function, $C(\tau)$, of the signal $x(t)$ has also the power-law decay form

$$|C(\tau)| \sim \tau^{-\gamma}, \tag{3}$$

where $\gamma = 1 - \beta$ is the correlation exponent related to the spectral exponent characterizing the power spectrum, and $|.|$ denotes the mathematical magnitude operator.

The case of signals of zero-autocorrelations corresponds to $\beta = 0$, $\gamma = 1$ (i.e., white noise). The case of signals with long-range positive autocorrelations corresponds to $\beta > 0$, $\gamma < 1$ (e.g., pink –bounded– and brown –unbounded– noise with $\beta = 1$, $\gamma = 0$ and $\beta = 2$, $\gamma = -1$, respectively); and the case $\beta < 0$, $\gamma > 1$ corresponds to signals with long-range negative autocorrelations (e.g., blue noise with $\beta = -1$, $\gamma = 2$). Fig. A8a shows the time series of the four kinds of simulated colored noise signals.

We also generated different profiles of filtered white noise signals in order to create linear noisy time series with a specific power structure. To this end, we applied 4th order Butterworth filters. Specifically, we used three different low pass filters upon an initial white noise signal with filter's cutoff frequency at 100, 50 and 10 Hz, in furtherance to construct different deflection frequency points at various positions in power. We also applied different bandpass and bandstop filters to the same initial white noise signal, namely, a bandpass and a bandstop filter between 10 and 100 Hz as well as a combination of the previous bandpass filter with two consecutive bandstop filters into the 10–20 Hz and the 40–60 Hz ranges, in the interest to built additional deflection points. Fig. A8b and c shows the time series of the six kinds of simulated filtered white noise signals.

Note that, all signals generated in this paragraph have a sampling rate of 1000 Hz.

2.1.1.2. Signals of nonlinear stochastic processes. In order to investigate how MSE curves are affected by nonlinearity and stochasticity, which are present in realistic signals like EEG (Nunez, 1989), we simulated nonlinear stochastic time series from a well-known dynamical system, the van der Pol oscillator (van der Pol and van der Mark, 1928). It is described by the following nonlinear equations:

$$\begin{cases} \dot{x} = y, \\ \dot{y} = -\omega_0^2 x + 2\mu\omega_0(1 - x^2)y + \eta(t) \end{cases} \tag{4}$$

where μ is the nonlinear damping strength parameter, $\omega_0 = 2\pi f_0$ is the natural pulsation of the system with f_0 being the natural frequency, and $\eta(t) = \sigma\xi(t)$ is a Gaussian process which satisfies

$$\langle \xi_t \rangle = 0, \quad \langle \xi_t \xi_0 \rangle = \delta(t - t_0). \tag{5}$$

Here $\delta(t)$ is the Dirac delta function and the notation $\langle . \rangle$ denotes the mathematical expectation operator. Note that the Heun method (Mannella, 2002) was used to integrate Eq. (4), with an integration step size of 10^{-4} ms. For $\mu = 0$, Eq. (4) represents a simple linear harmonic oscillator, for $\mu > 0$, the system enters a limit cycle, and for $\mu < 0$, Eq. (4) represents a stable equilibrium point (Argyris et al., 2015).

The generated signals shown in Fig. A8d–f resulted from the linear summation of 100 independent (stochastic) van der Pol oscillators with random initial conditions and oscillation frequencies from 1 to 100 Hz, with a step of 1 Hz. In the interest to study the effects of nonlinearity and stochasticity upon the compared metrics, different pairs of damping parameter, μ (i.e., $\mu = [-0.05, -0.5, 0.5, 1, 2]$), and noise intensity, σ (i.e., $\sigma = [0.01, 0.1, 500]$), have been selected. Thus, we created signals of a wide bandwidth (like the colored noise signals above), of multiple frequency rhythms (similar to biological signals), and of nonlinear and stochastic variable, although still far from the realistic biological complexity of signals such as EEG (Burke and de Paor, 2004; Nunez, 1989).

2.1.2. EEG data

In addition to the simulated signals, we also analyzed EEG data obtained from earlier recordings (Müller et al., 2009; Müller and Lindenberger, 2012). Briefly, 31 young adults (mean age = 22.7, standard deviation = 1.6, age range = 18.8–25.1 years, 14 females) and 28 older adults (mean age = 67.8, standard deviation = 3.0, age range = 63.9–74.5 years, 14 females) were involved in that study. For the purpose of the present work, only data from resting-state with eyes closed were used. Subjects were instructed to keep awake with their eyes closed for 1.5 min and relax. The results are averaged across artifact-free 10s epochs (with a sampling rate of 250 Hz) of all subjects within groups, and are represented in Fig. 7. Further details can be found in Sleimen-Malkoun et al. (2015). Moreover, for each data segment, we generated a surrogate one by random shuffling the segments' phases in the Fourier space according to Theiler et al. (1992) method. The resulting surrogate signals have the same linear properties, that is, autocorrelation and power spectrum, as the original data, but all other time dependencies, and therefore, all possible nonlinear autocorrelations originally present in the data are destroyed by the phase randomization.

Please note that the surrogate data analysis is not applied in the case of the simulated chaotic oscillators. Indeed, we contend of controlling the amount of nonlinearities by tuning the corresponding parameter in the generating signals. Adding a surrogate analysis would not offer more insights than what could be expected after taking into account the change in the degree of nonlinearity and the introduction of more randomness in the surrogates due to phase shuffling.

2.2. Data analysis

The simulated and experimental data were analyzed using Matlab R2012b (The MathWorks, Inc., Natick, MA, USA). All data were z-scored in preprocessing (in particular, simulated data were z-scored after generation).

2.2.1. Multiscale entropy analysis

MSE evaluates the structure of variability of signals across multiple temporal scales. Technically, MSE consists in computing Sample Entropy (Richman and Moorman, 2000) of the successively coarse-grained signals (Costa et al., 2002, 2005).

2.2.2. Coarse-graining procedure

Under this procedure, a time series $x = \{x_1 \ldots x_N\}$ is divided into segments that correspond to consecutive nonoverlapping time intervals of length τ_{SF}, where τ_{SF} represents the scale factor and takes integer values equal or greater than 1. Next, all data points in each segment are replaced by the average value of that segment, thus producing a new time series $y^{\tau_{\mathrm{SF}}} = \{y_j^{\tau_{\mathrm{SF}}}, j = 1 \ldots N/\tau_{\mathrm{SF}}\}$, the length of which is equal to that of the original time series, N, divided by the scale factor τ_{SF}. For scale 1, y^1 is simply the original time series.

$$y_j^{\tau_{\mathrm{SF}}} = \frac{1}{\tau_{\mathrm{SF}}} \sum_{i=(j-1)\tau_{\mathrm{SF}}+1}^{j\tau_{\mathrm{SF}}} x_i, \quad 1 \leqslant j \leqslant N/\tau_{\mathrm{SF}} \tag{6}$$

2.2.3. Sample entropy algorithm

Sample entropy (SampEn) quantifies the predictability or regularity of a time series (Richman and Moorman, 2000). It is defined as the negative natural logarithm of the conditional probability that two similar sequences of m consecutive data points in a time series of length N, within a given tolerance level r normalized to the standard deviation of the time series, will remain similar when the next point $(m+1)$ is also included in the sequence

$$\mathrm{SampEn}(m, r, N)(\tau_{\mathrm{SF}}) = -\log\left(\frac{\mathrm{P}^{m+1}(r)}{\mathrm{P}^{m}(r)}\right), \tag{7}$$

where the quantity $\mathrm{P}^{(m+1)}(r)/\mathrm{P}^{(m)}(r)$ is the conditional probability described above.

Following several studies employing MSE across imaging modalities (Lippe et al., 2009; McIntosh et al., 2008, 2014; Misic et al., 2011; Sleimen-Malkoun et al., 2015), we set the pattern length, m, to 2 and the tolerance criterion, r, to 0.5 throughout all computations of MSE performed in this study. Furthermore, to ensure the robustness of the analysis, and unless otherwise specified, we use approximately 30,000 data samples in each simulation and coarse-grained them up to scale 100. Please note that SampEn requires at least 10^m to 20^m data points for a robust calculation (Richman and Moorman, 2000). Our shortest coarse-grained time series include 300 data points.

2.2.4. Power spectrum analysis

Previous studies suggest that changes in MSE curves are related to changes in the frequency content of the underlying signal (McIntosh et al., 2014; Mizuno et al., 2010; Vakorin et al., 2011). With the aim of exploring how specific spectral components are expressed in the MSE curve, power spectrum (PS) analysis is performed by means of the Welch's averaged modified periodogram method (Welch, 1967) with a Hamming window and without overlap, using the Matlab pwelch function.

Note that the PS reflects reliably only linear autocorrelations in a signal, as the spectral density is a Fourier transform pair of the autocorrelation function (Wiener–Khinchin theorem, Wiener, 1930), which by definition evaluates linear autocorrelations. However, possible nonlinearities (e.g., as those of a nonlinear van der Pol oscillator that deviates from the harmonic oscillator) can still express themselves as harmonic peaks.

2.2.5. Linear autocorrelation analysis

For the purpose of investigating whether the signal's linear autocorrelation properties are captured in the MSE curve, we use a new method, namely the multiscale root-mean-square-successive-difference (MRMSSD) analysis. The MRMSSD is an extension of the previously introduced root-mean-square-successive-difference (RMSSD) algorithm that quantifies the average magnitude of variability of changes between two consecutive time points (von Neumann et al., 1941). In this respect, MRMSSD can be considered as less integrative than MSE.

The MRMSSD algorithm consists of two steps: (1) a coarse-graining procedure, identical to the one used in the MSE algorithm, is used to derive the representation of the system's dynamics at different time scales; (2) the RMSSD algorithm is used to quantify the amplitude variability of the coarse-grained time series at each scale factor. As its name implies, RMSSD is calculated as

$$\mathrm{RMSSD} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}(x_i - x_{i+1})^2}, \tag{8}$$

where N is the number of successive distinct pairs of sampling time points x_i and x_{i+1}. From the relationship between this metric and the temporal autocovariance function, Cov (Wasserman, 2013),

$$\mathrm{RMSSD}^2 = 2(\sigma^2 - \mathrm{Cov}), \tag{9}$$

where σ^2 is the variance of the time series, we can draw a relationship between RMSSD and the temporal autocorrelation function, C,

$$\mathrm{RMSSD} = \sqrt{2\sigma^2(1 - \mathrm{C})}, \tag{10}$$

where $\mathrm{C} = \mathrm{Cov}/\sigma^2$. Hence, RMSSD can be used to evaluate linear autocorrelations, and we can compare its multiscale version with MSE that shares the same procedure to focus on different scales.

Note though that, in a similar fashion to spectral analysis, possible nonlinear terms of a dynamical process, such as the ones contained in the here tested van der Pol oscillators, can still express themselves via their projection to the corresponding linear space (e.g., a sigmoidal autocorrelation would still be detected as a positive linear autocorrelation).

In order to examine the relationship between the metrics in time and frequency domain, we provide a correspondence between scale factor τ_{SF} (number of iteration of the coarse-graining) and frequency (in Hz). In that regard, the highest frequency ($f_{\mathcal{N}}$) in the signal is calculated following the Nyquist-Shannon' s sampling theorem as $f_{\mathcal{N}} = f_{\mathcal{S}}/2$, where $f_{\mathcal{S}}$ is the sampling frequency of the signal

(Shannon, 1949). Hence, the relationship between frequency and scale factor is given by

$$f_{\mathcal{N},\tau_{SF}} = \frac{f_{\mathcal{S},1}}{2 \times \tau_{SF}}, \tag{11}$$

where $f_{\mathcal{S},1}$ corresponds to the sampling frequency at time scale 1, i.e., the one of the original time series. Additionally, this correspondence is made more straightforward by converting the scale factor to time scale, τ

$$\tau = \frac{\tau_{SF}}{f_{\mathcal{S},1}} \times 1000. \tag{12}$$

The factor 1000 is for expressing everything in ms. To capture the overall changes in the distribution of structure and amplitude variability, the log–log graphs are presented for each time series' analysis.

3. Results

3.1. Simulation results

3.1.1. Linking linear temporal correlations to the scaling of MSE curves.

First, we study the scaling properties of MSE curves for four kinds of colored noise signals that are characterized by different linear temporal autocorrelations. The four signals, that is white, pink, brown and blue noise, were constructed to exhibit a PS that scales with frequency f as $f^{-\beta}$, where β is the spectral exponent that equals to 0, 1, 2 and −1, respectively (see Fig. 1a, where the scaling on the log–log plot is linear with a slope of approximately $-\beta$). As a result, the amplitude of the autocorrelation function of these signals should scale with time scale τ as $\tau^{-\gamma}$, where $\gamma = 1 - \beta$ (i.e., γ equals to 1, 0, −1 and 2, respectively; see Section 2). As it is shown in Fig. 1b, the logarithm of MRMSSD linearly scales with the logarithm of τ, with the slope being very close to the value $-\gamma/2$ (i.e., −0.5, 0, 0.5 and −1, for the same order of signals), which was expected due to the square root relationship between RMSSD and the autocorrelation function (see Eq. (10)). Concerning the MSE, it also linearly scales with time scale τ with an exponent of approximately $-\gamma/2$, for pink and brown noise that contain positive correlations. This is however not the case for white and blue noise that contain zero and negative correlations, respectively (see Fig. 1d for MSE vs. τ in log-log space). Moreover, we see that pink noise, which is the bounded signal, has equally high SampEn values across scales, while brown noise, the unbounded signal, presents increasing values with increasing scales.

Thus, we conclude that in the absence of any nonlinearities, MSE evaluates long-range linear autocorrelations, and scales with a power-law only for signals with positive correlations.

3.1.2. Linking power spectra to MSE curves

In the following, we investigate how the structure of the PS of a signal is reflected in the profile of its MSE curve, still in presence of only linear autocorrelations (i.e., for a signal of linear stochastic processes). In particular, we are interested in determining how deflection frequency points, created in the PS via the application of different filters on an initially flat (white noise) spectrum, yield distinct peaks in the MSE and MRMSSD curve.

First, we study the effects of the coarse-graining procedure upon the PS of the white noise signal after a low pass filter is applied with a cutoff frequency (Wc) of 100 Hz (Fig. 2). Coarse-graining acts as a low pass filter operation in the frequency domain (Govindan et al., 2007; Kaffashi et al., 2008; Valencia et al., 2009); hence looking at the resulting PS curves of the successive coarse-grained signal (small panels in Fig. 2), we observe that the signal at fine time scales (e.g., τ = 1 and 5 ms) contains high frequencies in addition to low frequencies, whereas at coarser time scales (e.g., τ = 50 ms) only low frequencies remain. Moreover, the respective MSE and MRMSSD curve increase up to approximately a time scale of 5 ms, and decrease for longer scales, thus inducing a global peak at 5 ms that corresponds, in frequency, to the upper limit of the bandwidth of the signal (i.e., 100 Hz). Indeed, using the relationship between time scale and frequency (see Eqs. (11) and (12)), we obtain

$$\tau_{SF} = \frac{f_{\mathcal{S},1}(= 1000\,\text{Hz}) \times \tau(= 5\,\text{ms})}{1000} = 5$$

$$f_{\mathcal{N},\tau_{SF}} = \frac{f_{\mathcal{S},1}}{2 \times \tau_{SF}} = 100\,\text{Hz}.$$

The fact that the MSE and MRMSSD peak relate to the bandwidth of the signal is better demonstrated in the next results, in which the position of the peak is moved to the left by applying two different low pass filters to the same initial white noise signal (Fig. 3). Specifically, we generated white noise signals with a bandwidth up to 50 and 10 Hz. Thus, this yields MSE and MRMSSD curve with peaks at approximately 10 and 50 ms, respectively.

Then, in order to create additional deflection frequency points in the PS, we increased the complexity of its structure by applying different bandpass and bandstop filters upon the initial white noise signal. The first signal (Fig. 4, in blue) is generated with a bandpass filter in the range 10–100 Hz, which results in an inverted-U curve with a tail for both MSE and MRMSSD, in contrast with the peak-less fast decaying curve of the full spectrum of the white noise signal (i.e., in the range 1–500 Hz shown in grey). The second signal (Fig. 4, in orange) is obtained by combining the latter bandpass filter with two bandstop filters in the ranges 10–20 Hz and 40–60 Hz, respectively. By doing so, a two-peak structure is observed for the MRMSSD (at τ = 11 and 42 ms, i.e., 45 and 12 Hz, respectively) corresponding to a flattening of the MSE at similar time scales (τ = 12 and 47 ms, i.e., 42 and 11 Hz, respectively). In both the above cases, the global maximum at τ = 5 ms, corresponding to the cutoff frequency of the bandpass filter at 100 Hz, remains. Finally, a third signal is obtained by applying a bandstop filter in the range of 10–100 Hz upon the full spectrum of white noise signal (Fig. 4, in green). The structural variability of this signal shows a fast drop with low MRMSSD and MSE values in the range of 5–50 ms, which correspond to the filter's cutoff frequency interval. MSE remains almost flat for that region, whereas an additional local peak appears in the MRMSSD curve for τ = 50 ms approximately (i.e., for 10 Hz).

In sum, our investigations with filtered white noise signals reveal how changes in the structure of the PS are reflected in the curves of the MSE and MRMSSD, given the downsampling effects of the coarse-graining procedure. When the frequency content of the signal is bounded, MSE and MRMSSD curves present a single global peak that corresponds to the upper frequency of the signal's bandwidth. When power is removed for specific ranges through filtering, it leads to an inflexion in MRMSSD and MSE curve at the corresponding time scales. Conversely, for the intact parts of the spectrum, both measures show a decrease with a slope that is characteristic for white noise. Whereas MSE curves tend to flatten for time scales that correspond to frequency ranges where filters have been applied, MRMSSD curves increase and show additional local peaks. We conclude that MSE and MRMSSD – in a signal presenting only linear autocorrelations – track all the changes in the PS at the corresponding time scale ranges, but their actual values at each time scale integrate power values across the whole spectrum that survive the coarse-graining at that scale, i.e., MSE and MRMSSD reflect the PS in an integrative manner. This is something to be expected for measures of linear autocorrelations, given the Wiener–Khinchin theorem, which states that the autocorrelation function of a wide-sense-stationary random process has a spectral decomposition given by the PS of that process (Wiener, 1930).

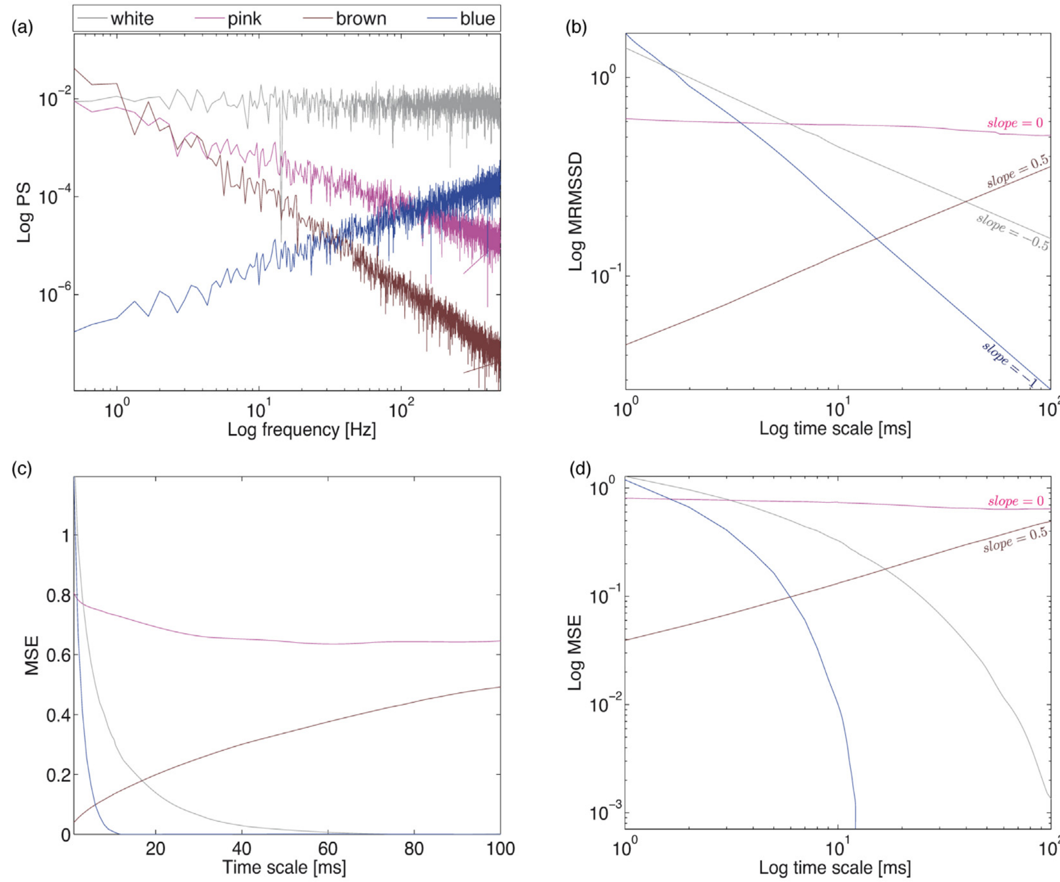

Fig. 1. Linking linear temporal correlations to the scaling of MSE curves in simulated colored noise signals: (a) power spectrum, (b) multiscale root-mean-square-successive-difference, (c) and (d) multiscale entropy in linear and logarithmic scale, respectively. Line colors correspond to the color of noise (except for white noise that is shown in grey). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

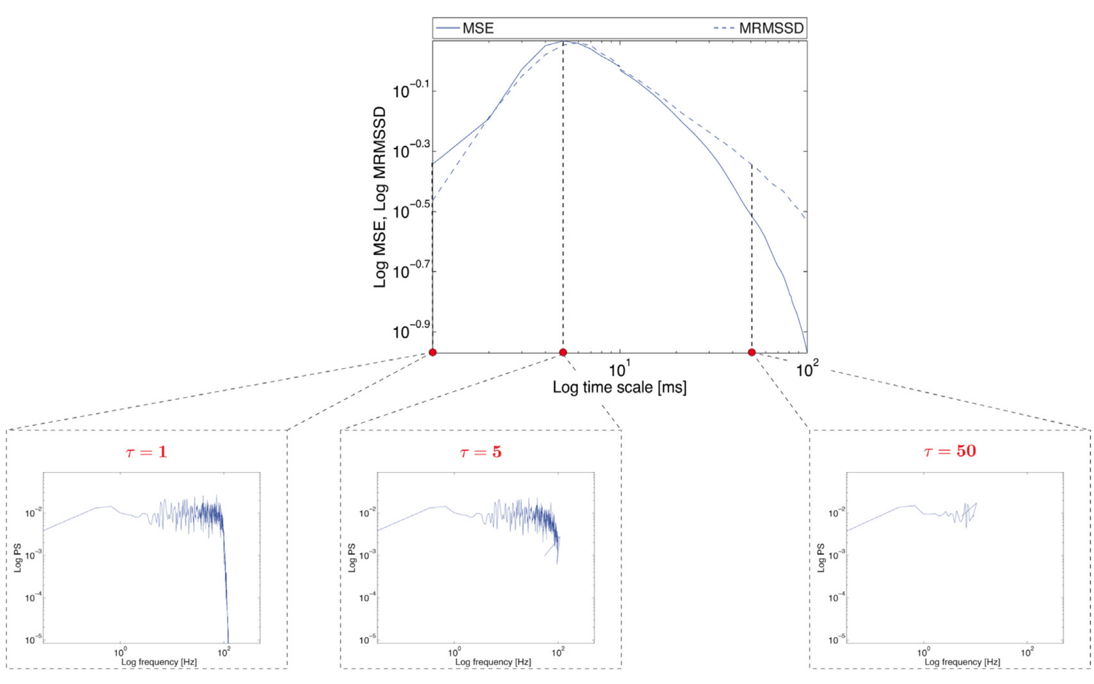

Fig. 2. Coarse-graining effects on the structural variability profiles of low pass filtered white noise signal. The curves are presented in log–log space, with multiscale entropy in solid line and multiscale root-mean-square-successive-difference in dashed line. The small panels depict the power spectrum of the coarse-grained signal at specific scales, that is, 1, 5 and 50 ms.

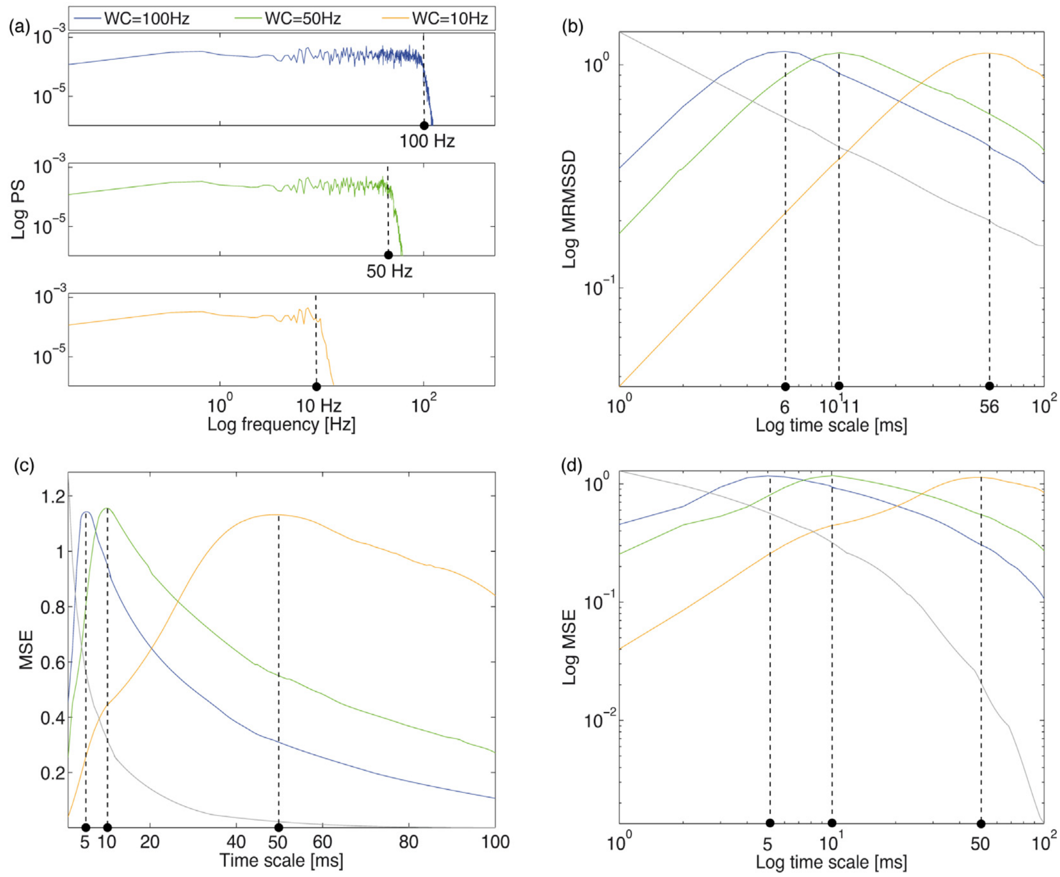

Fig. 3. Relationship between the bandwidth of a signal and the peak structure of the structural variability curves of low-pass filtered white noise signals: (a) power spectrum, (b) multiscale root-mean-square-successive-difference, (c) and (d) multiscale entropy in linear and logarithmic scale, respectively. Curve colors correspond in blue to filter's cutoff frequency of Wc = 100 Hz, in green of Wc = 50 Hz, and in orange of Wc = 10 Hz; the grey curve represents the original white noise signal. The black dashed lines illustrate the deflection points in frequency and their counterpart in time scale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

3.1.3. Comparing the effects of nonlinearity and stochasticity on MSE curves

Having, till now, established the relationship between MSE and the other metrics, in cases of random signals and in the presence of only linear autocorrelations, we now compare the effects of nonlinearity and of interactions between deterministic and stochastic components. (Note that, as explained above, PS and MRMSSD analysis cannot reliably evaluate the effects due to nonlinear dynamics, even though some changes may be still indirectly detected as the presence of harmonics, for example).

In order to get a little closer to biological complexity, we generated broadband multi-rhythmic signals by simulating, and superimposing, 100 van der Pol oscillator time series. These oscillators have natural frequencies from 1 to 100 Hz, with a step of 1 Hz, and are of equal amplitudes. We generated versions of this signal in the oscillatory regime of varying nonlinearity (μ = [0.5, 1, 2]), and with or without additive dynamic noise (σ = 0 and 500) (see Fig. A8d and e). In the deterministic case (Fig. 5, in solid curves), we observe that increasing the nonlinearity has strong effects on MSE curves leading to higher MSE values. We should also note that MSE curves have apparent artifact peaks at coarse scales. Indeed, the coarse-graining procedure, which is equivalent to the application of a finite-impulse response (FIR) filter to the original time series, and to the downsampling of the filtered time series with a factor τ (Nikulin and Brismar, 2004; Valencia et al., 2009), does not eliminate the fast temporal scales above the Nyquist's frequency $f_{\mathcal{N}}(\tau)$. Thus, the subsequent downsampling procedure produces aliasing generating spurious oscillations in the frequency range from 0 to $f_{\mathcal{N}}(\tau)$, in particular if there is high-frequency content in the signal, as it is the case here (Oppenheim and Schafer, 1975).

Turning now to the stochastic oscillators (Fig. 5, in dashed curves), we observe that there are no more evident artifact peaks indicating that these artifacts are stronger for signals that are more regular and have clearer spectral peaks (i.e., the deterministic signals). Moreover, MSE increases across scales with stochasticity, something to be expected for a metric that evaluates irregularity in the signal. Nevertheless, regarding the effects of nonlinearities, they are much less evident and, counterintuitively, of opposite direction compared to the deterministic signals. Namely, MSE slightly reduces with increasing nonlinearity (except at fine scales due to the effects of high frequency harmonics). This latter effect could be due to a more complicated interaction between nonlinearity, stochasticity, and the total damping strength of the system (but see also the next paragraph on the damped oscillators).

In any case, and in contrast to MSE, MRMSSD curves show little modulation with either nonlinearity or stochasticity, except for the fine scales due to the effects of high frequency harmonics, while, at the same time, they are also more robust to downsampling artifacts. In particular, MRMSSD decays with a power-law signature of approximately −0.5 at coarse scales (i.e., at τ = 5 ms corresponding to the signal's bandwidth), indicating that the correlation exponent is approximately 1, a value to be expected for signals with flat – white noise-like – spectrum. Indeed, the spectral exponent is approximately 0 (see Section 3.1.1). Finally note that the increase of nonlinearity for the deterministic case results in a smoothing of the harmonic structure of the PS.

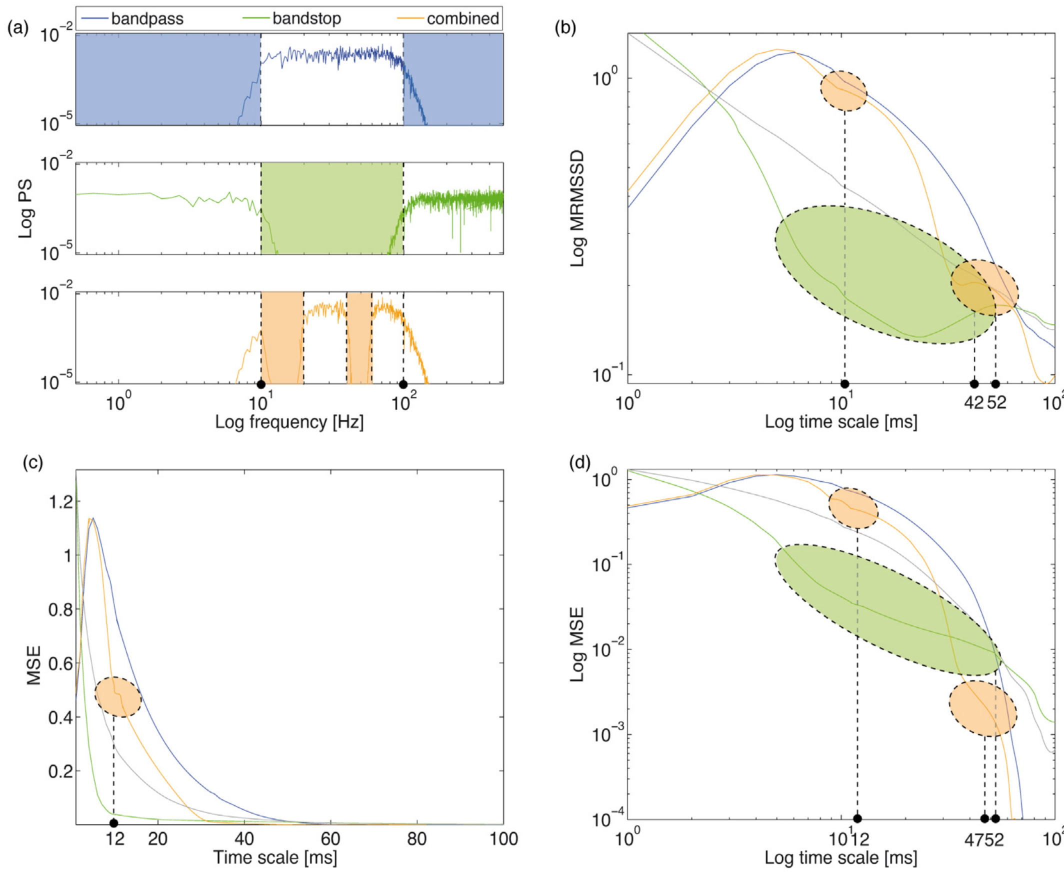

Fig. 4. Relationship between power structure and structural variability profiles of filtered white noise signals: (a) power spectrum, (b) multiscale root-mean-square-successive-difference, (c) and (d) multiscale entropy in linear and logarithmic scale, respectively. Curve colors correspond in blue to the bandpass filtered signal, in green to the bandstop filtered signal, and in orange to the combined filtered signal; the grey curve represents the original white noise signal. Shade areas illustrate the filter's cutoff intervals, and the black dashed lines are the deflection points in power and their counterpart in time scale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

At this stage, one can conclude that in the case of a limit cycle, the PS and MRMSSD, which specifically captures linear effects, can show nonlinearities via harmonics. On the other hand, MSE is clearly more sensitive to nonlinearity, although it is subject to artifacts due to downsampling effects, especially in presence of high frequencies, weak nonlinearities, and high stochasticity. This is an important issue to take into account when analyzing highly regular signals at long scales.

Next, we compare the effects of nonlinearity and stochasticity on the tested metrics for a different kind of system, namely, for stochastically damped oscillations. As done previously, we constructed a signal based on 100 superimposed stochastic van der Pol oscillators with different, here negative, damping parameters (μ = −0.05 and −0.5), that is, in the stable equilibrium state, and different noise intensities (σ = 0.01 and 0.1). This regime gives rise to noisy damped oscillations with "constant" drift (although all parameters are autonomous, it still depends on the state variable) and diffusion terms (Fig. A8f).

In all the cases, the noise (i.e., the diffusion term) is of the same intensity at each oscillator; however, the determistic part, which corresponds to the drift term, involves terms proportional to $\mu\omega_0$ and ω_0^2 (see Eq. (4)), which have different influences on the dynamics depending on how far the system is from the equilibrium point (here, the origin). Therefore, the damping time is inversely proportional to $\mu\omega_0 + \omega_0^2$ and will be much shorter for high frequencies. Consequently, the PS should show a steep decay for all signals. As it is shown in Fig. 6a, we observe that the PS decrease linearly with a scaling of approximately −3, across all frequencies, and for all signals. As a result, both MSE and MRMSSD curves show an increase across time scales with a power-law signature of approximately 1 (see Fig. 6b–d). This indicates that the correlation exponent γ is approximately −2, a value to be expected from the PS's scaling. Moreover, we observe that increasing nonlinearity, as well as stochasticity, leads to higher MSE values. In contrast, MRMSSD does not seem to be modulated by the level of stochasticity. Regarding the effects of nonlinearities, they are in the opposite direction for MSE and MRMSSD. In particular, while MSE increases, MRMSSD reduces with increasing nonlinearity at coarse scales. Note that we can also observe some artifact peaks at coarse scales in the MSE curves.

Thus, in the case of a stable equilibrium state, it seems there is an interaction between nonlinearity and stochasticity, since a minimum amount of noise is needed for the system to explore its state space and its nonlinear dynamics.

3.2. Experimental EEG results

Finally, using the intuitions obtained from the above results, we look back to some of the results reported in Sleimen-Malkoun et al. (2015) examining the effects of aging on the variability of EEG signals. We reproduced the group average PS and MSE curve for young and old adults at rest with eyes closed. Compared to the analysis in this study, we extended the computation of MSE to longer scales and computed MRMSSD for comparison and we additionally repeated the whole analysis for surrogate data, which were generated by phase randomization in the Fourier space (Theiler et al.,

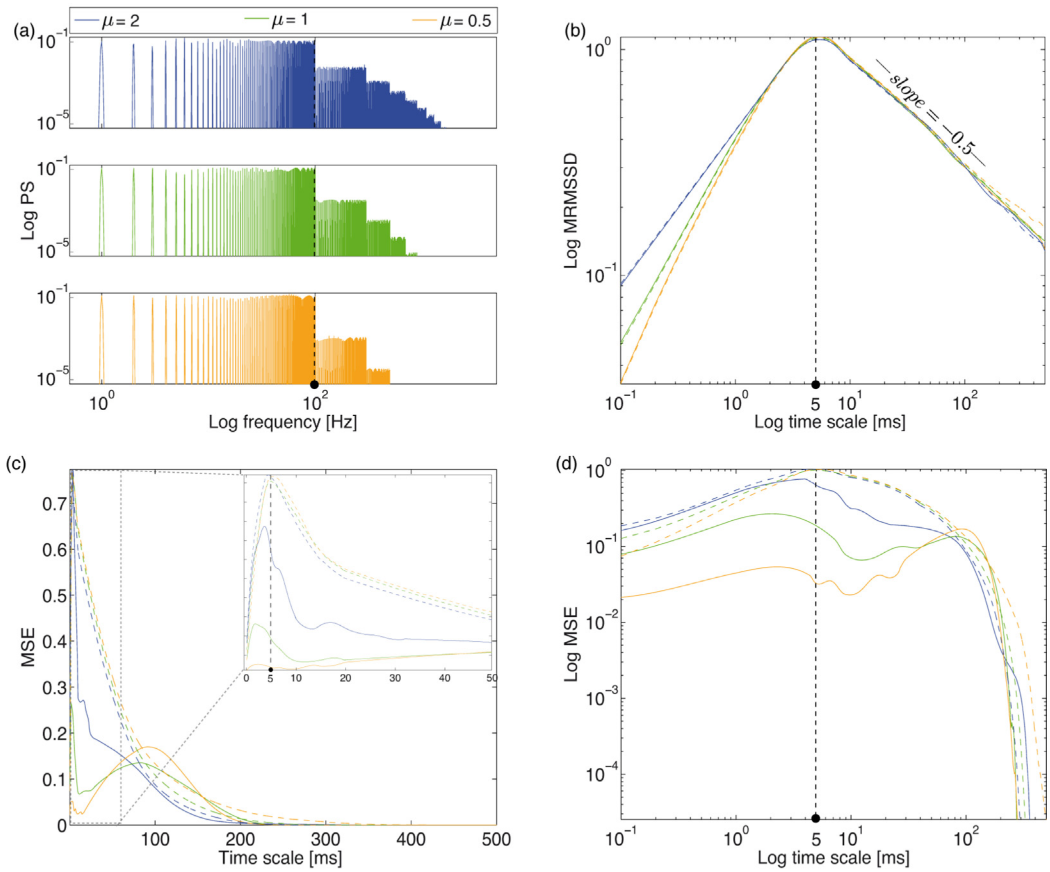

Fig. 5. Tracking effects of the amount of nonlinearity and stochasticity in a signal on the tested metrics from simulated 100 superimposed van der Pol oscillators in the oscillatory regime: (a) power spectrum, (b) multiscale root-mean-square-successive-difference, (c) and (d) multiscale entropy in linear and logarithmic scale, respectively. Solid curves represent the deterministic signals and the dashed ones to their stochastic version. Curve colors correspond in blue to $\mu = 2$, green to $\mu = 1$, and orange to $\mu = 0.5$. The black dashed lines illustrate the upper limit frequency in power and their counterpart in time scale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

1992). Group differences (young vs. old adults) can be inspected in Fig. 7, where the mean values and standard error intervals of the Cz electrode are shown, both for the original (Fig. 7, dark color curves) and surrogate data (Fig. 7, light color curves).

We observe that young adults have more power in the theta and alpha bands than old adults, whereas the results are reversed for higher frequencies. Such a crossing point is reproduced also for the MSE and MRMSSD curves around the time scale $\tau = 24$ ms. These results are in accordance with Sleimen-Malkoun et al. (2015) findings as well as previous studies (McIntosh et al., 2008, 2014; Vakorin et al., 2011; Yang et al., 2013). Moreover, the global peak of the MSE and MRMSSD curves corresponds to a slightly coarser scale for young adults (i.e., at $\tau = 36$ ms), whereas it is at 20 ms for older adults. Regarding the surrogate data, the curves follow very closely those obtained from the original data. Namely, while the original curves are indistiguishable from their surrogates with the MRSSD analysis, the latter show slightly higher MSE values, and can only be distinguished from them around the global peak of the curves.[2] Higher entropy in surrogates can be expected as the phase randomization induces more unpredictability in the signal (see also Vakorin and McIntosh, 2012).

These results taken together, we can confirm that EEG signals of young adults, at rest, are characterized by a slower time constant than the older adults. In addition, the surrogate data analysis, together with the observation that group separation is clearer through in the MRMSSD curves than the MSE ones, speak in favor of the effects being mainly due to the linear autocorrelations.

4. Discussion

Introduced as a measure of complexity of physiological signals (Costa et al., 2002, 2005), MSE is thought to evaluate the presence of long-range correlations in the behavior under scrutiny by taking into consideration the multiple time scales, on which the overall dynamics operates. The main interest of its application to study brain dynamics has been the possibility of explicitly linking these time scales to local and global processing in neural populations (McIntosh et al., 2014; Mizuno et al., 2010; Vakorin et al., 2011), as well as inferring a time scale dependent relation between complexity and neural connectivity (McDonough and Nashiro, 2014; Nakagawa et al., 2013).

[2] In Appendix B, the reader can find an explorative partial least squares analysis across all channels, showing that for the MSE analysis, there is a significant latent variable in the data, which contrasts original and surrogate data. However, this latent variable accounts for only 10.5% of the covariance. The rest is explained by a latent variable corresponding mainly to the contrast between young and old adults (i.e., group effect). In addition, there is no significant interaction between these two effects. Therefore, it can be concluded that the nonlinearities that exist in the data are weak and hardly influence the age group differences. Moreover, MRMSSD showed only one significant latent variable, explaining almost all covariance in the data along a contrast mainly between young and old adults, i.e., it could not detect any effect of possible nonlinearities in the data.

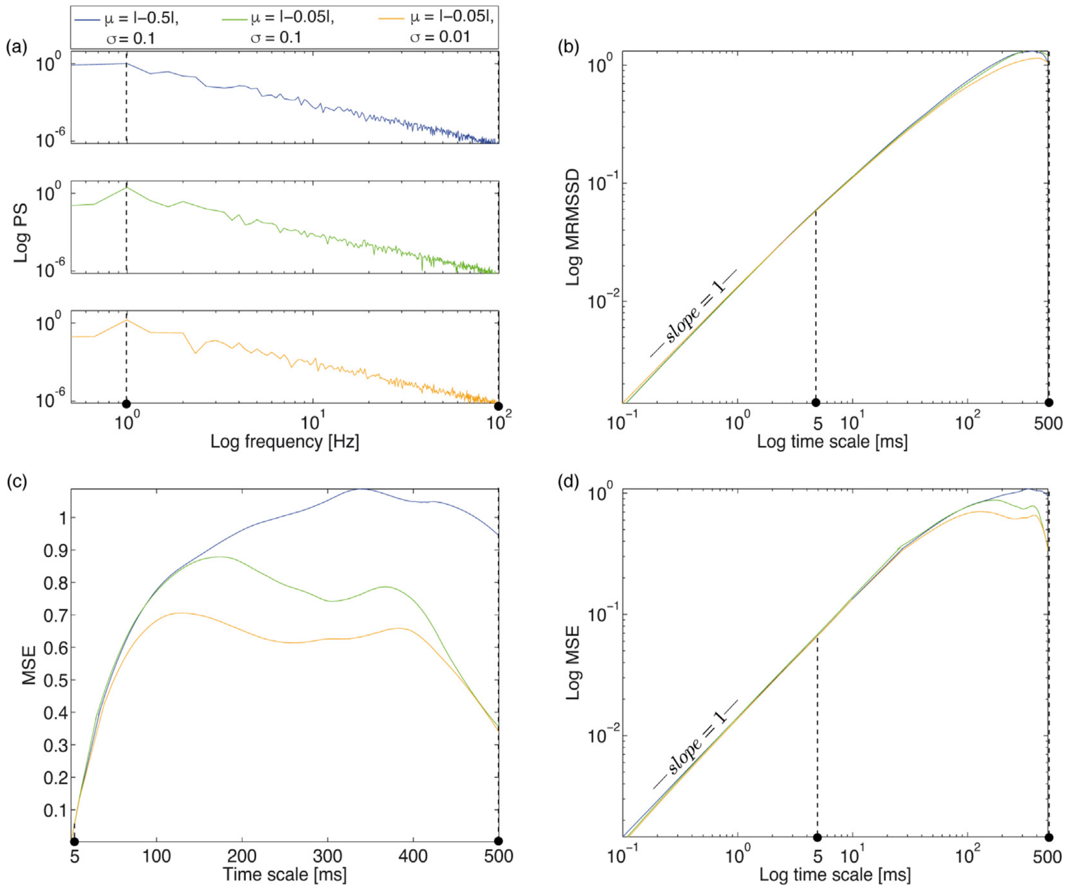

Fig. 6. Effects of nonlinearity and stochasticity on the tested metrics of 100 superimposed stochastic van der Pol oscillators in stable equilibrium state: (a) power spectrum, (b) multiscale root-mean-square-successive-difference, (c) and (d) multiscale entropy in linear and logarithmic scale, respectively. Curve colors correspond in blue to $\mu = |-0.5|$ and $\sigma = 0.1$, in green to $\mu = |-0.05|$ and $\sigma = 0.1$, and in orange to $\mu = |-0.05|$ and $\sigma = 0.01$. The black dashed lines represent the bandwidth of the signals in power and their counterpart in time scale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Here, we systematically investigated how the frequency content, its scaling and its structure are expressed in the MSE profile. Next, we demonstrated how linear and nonlinear autocorrelations, as well as stochastic components, affect MSE curves. A better intuition was gained by inspecting jointly the power spectrum and the multiscale root-mean-square-successive-difference of the studied signals. We introduced this latter metric as a proxy of the autocorrelation function to study the presence of linear long-range correlations. We successively inspected simulated signals that are characterized with different power structures, linear and nonlinear autocorrelations properties as well as different deterministic and stochastic components. Finally, we analyzed experimental EEG data to illustrate the application and interpretation of MSE with brain signals.

4.1. MSE and linear autocorrelations

First, we showed that the complexity, as evaluated by the MSE algorithm, is closely related to the autocorrelation structure across the time scales of the signal. This was demonstrated by using colored noise signals, for which linear autocorrelations and spectra scale with distinct decay power-laws. We found that MSE scaled with the same power-law as MRMSSD for signals with only positive autocorrelations (e.g., pink and brown noise). Conversely, it presented a fast drop with increasing time scale, but without a power-law, for signals that are either uncorrelated (i.e., white noise) or negatively correlated (e.g., blue noise).

Then, we studied the effect of the coarse-graining procedure through which the MSE algorithm extracts the different time scales from the original signal. Practically, the coarse-graining consists in a downsampling operation (cf. also Govindan et al., 2007; Kaffashi et al., 2008; Valencia et al., 2009), which progressively removes fast oscillating activity from the signal, so that, as the time scale increases, it is essentially the entropy of the slower activity (i.e., the lower frequencies) that is evaluated. In contrast, the finer scales comprise both fast and slow oscillations.

A prominent feature of MSE curves is the presence, or not, of a peak at a defined time scale. In that regard, we illustrated how the corresponding time scale is determined by the upper limit of the signal' s bandwidth. We showed as well how the profile of the MSE curve is shaped by the power content and structure of the signal by altering the PS of the white noise signal. Namely, by applying different bandstop and bandpass filters resulted in a flattening of MSE curve for the parts of the spectra where power was removed. In this particular case with no nonlinear autocorrelations, the MRMSSD curve was affected similarly, except that bandstop filters created therein additional peaks. Thus, it appears that MSE offers a smoother representation of the peak structure of the PS than MRMSSD. In any case, both measures can be considered as an integrative representation of the PS, in the sense that their value at each time scale depends on the whole part of the spectrum that remains intact after coarse-graining. It is noticeable, however, that MRMSSD as a measure of short-term variation (successive difference) reflects mainly the faster remaining components of the signal, and hence is less integrative than MSE.

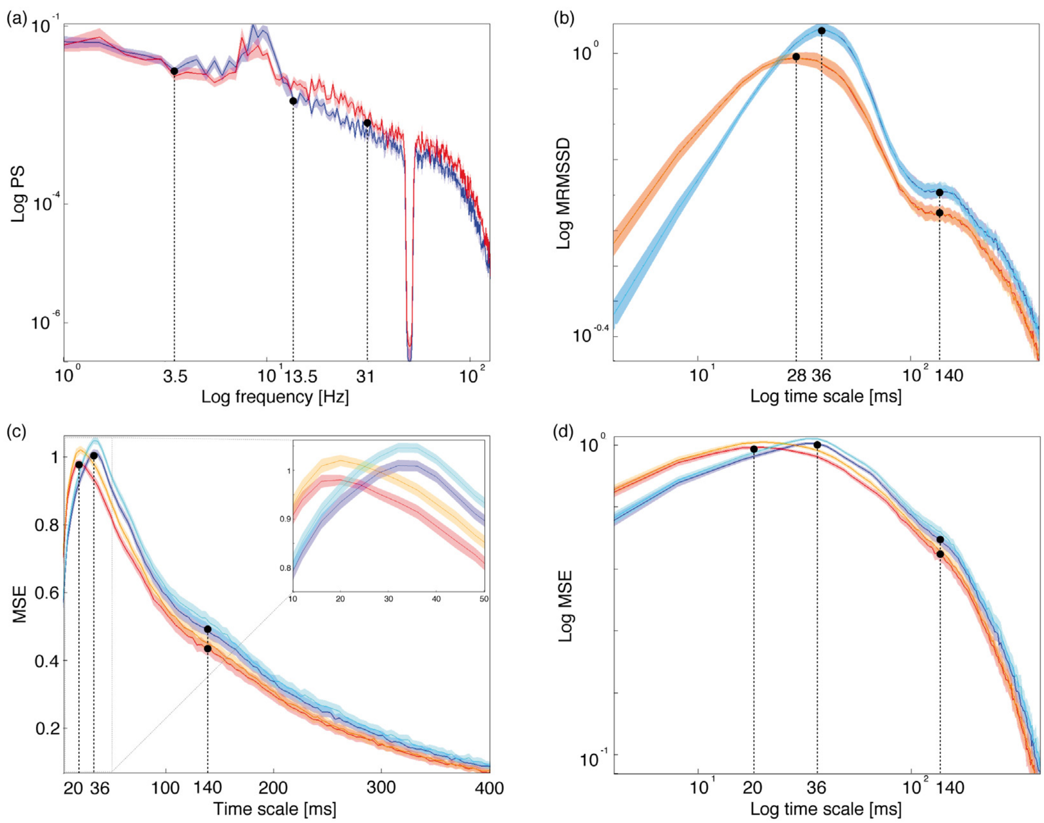

Fig. 7. Application of variability measures to study age group differences in EEG dynamics. Mean groups results of channel Cz for the resting state condition: (a) power spectrum, (b) multiscale root-mean-square-successive-difference, (c) and (d) multiscale entropy curves in linear and logarithmic scale, respectively. Young adults' curves are depicted in blue and those of older adults in red, and their surrogates are depicted in light blue and light red, respectively. Areas of faded colors represent the standard error intervals. The black dashed lines illustrate the deflection points in power and their counterpart in time scale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

4.2. MSE and nonlinearities

Regarding nonlinear autocorrelations and their interactions with stochastic dynamical components, we showed that MSE is much more sensitive to these factors than MRMSSD or PS analysis. Increasing the nonlinearity parameter of 100 superimposed van der Pol oscillators of a broadband multi-peaked spectrum modified the PS mainly via the addition of harmonics. At the same time, the effects were only weakly expressed in the MRMSSD curves. Instead, MSE differentiated signals of weak nonlinearity from those with strong nonlinearity, and deterministic from stochastic ones. However, we showed that MSE was more subject to downsampling artifacts in the deterministic signals. A clearer insight on the effects of nonlinearity, stochasticity and their interaction, can be gained through future analyses using different nonlinear systems.

4.3. MSE and age-related differences in EEG signals

Finally, we presented the example of a well-known result in the aging literature about group differences between young and old adults in terms of their EEG variability, as it can be seen via their PS and MSE curves. The weak nonlinearities present in the data hardly influenced the differences between young and old adults.

We reconsidered these results using the aforementioned findings. We showed that, MSE results confirm what can be seen through the PS and the MRMSSD analysis in terms of group differences. Although, MSE seems to be a smoother and more integrative representation of the frequency content and structure of the signal, MRMSSD, which evaluates only linear autocorrelations, offers a clearer distinction between the two groups. Accordingly, one can conclude that the age group differences, that have been reported in the literature using MSE analysis (McIntosh et al., 2014; Sleimen-Malkoun et al., 2015), are mostly due to changes in linear stochastic processes underlying EEG signals. This was confirmed by our surrogate data analysis (see also McIntosh et al., 2008; Vakorin and McIntosh, 2012), through which phases were randomized in Fourier space, destroying thereby nonlinear autocorrelations while preserving the linear ones. As compared to the real data, the results of this analysis showed a small increase of MSE values that was mostly visible around the global peak. Neither the general shape of MSE curves, nor the peak positions or crossing points between age groups were substantially different for the surrogate data. However, the link between neurophysiological interpretation (short/long-range communication, or else) and linearity/nonlinearity, which could be an interesting aspect to explore in the context of age-related alterations, remains unclear. To the best of our knowledge, there is no obvious systematic or lawful relationship between communication range and linearity/nonlinearity. One may evoke the nonlinearity present in neural dynamics (such as threshold behavior), but this topic is out of the scope of the current work.

From a different perspective, the inspection of the age-induced changes in MSE and MRMSSD profiles, which showed a crossing as well as a peak shift, suggests that, as compared to their younger counterparts, elderly have a wider bandwidth in their EEG activity, in which faster oscillations lead the dynamics and yield higher complexity at finer scales. If one considers that fast activity is determined by shorter neural connections on the basis of anti-correlations between spectral and spatial frequencies in EEG (Nunez, 1989), then the elderly brain is indeed characterized by a more local, segregated information processing (McIntosh et al., 2014; Mizuno et al., 2010; Vakorin et al., 2011). In that regard, the inclusion of additional evidence, such as the correlation of local effects of MSE at fine (respectively, coarse) scales with functional connectivity at fast (respectively, slow) frequencies in Ghanbari et al. (2015) as well as with local (respectively, distributed) entropy in McIntosh et al. (2014), provide stronger arguments.

4.4. Methodological concerns

Our results support the idea that MSE is a computationally efficient way to evaluate long-range correlations in biological signals. MRMSSD, although easier to compute and more sensitive to the structure of the PS, evaluates only linear autocorrelations and its use is recommended in the case of very weak nonlinearities and/or for signals of a very short length, in which MSE cannot be robustly calculated. PS, on the other hand, neither estimates nonlinear autocorrelations, nor offers an integrative representation that is generally desirable when comparing between groups and conditions. In fact, the present study was partly motivated by the contradicting evidence in the literature about the degree of similarity and/or complementarity of MSE and PS analysis. For instance in Catarino et al. (2011), Heisz et al. (2012), Takahashi et al. (2009) and Ueno et al. (2015), some significant effects were only found with MSE, and in Misic et al. (2011), the effects revealed by MSE were very different from those of spectral analysis. However, in the majority of studies, similar or slightly complementary effects were observed (Lippe et al., 2009; McIntosh et al., 2008, 2014; Misic et al., 2011; Mizuno et al., 2010; Sleimen-Malkoun et al., 2015). The surrogate tests that we used here (see also McIntosh et al., 2008) indicate that MSE, unlike power distribution, is sensitive to time dependencies and nonlinearities. Our results also strengthen the idea of complementarity between MSE and PS analysis and stress the fact that their joint use is a recommended practice to avoid misinterpretations. In addition, although assuming stationarity, we showed that the sensitivity of MSE to nonlinearity is not the only possible source of such complementarity, since the coarse-graining and the SampEn algorithm themselves make MSE a more integrative representation of the PS. Thus, unless PS shows no effect at all, additional tests like with surrogate data or a contextual knowledge are needed to properly distinguish the contribution of nonlinear deterministic components to the observed results. In that regard, we show that MRMSSD can also play a helpful role by using the same coarse-graining procedure as MSE, but evaluating only linear autocorrelations. This method provides an additional test to detect or reject the contribution of effects of a nonlinear nature. In the analyzed exemplary EEG data presented here, MRMSSD not only mirrored the result of MSE, but also offered a clear and reliable (see Appendix B) age groups separation. This indicated that these differences could be attributed mainly to linear dynamics, which was further confirmed by the surrogate data analysis.

4.5. Conclusion

Overall it can be concluded that, although not straightforward, MSE curves obtained from brain signals can be linked to its power content in terms of low and high frequencies, as well as provide information about both linear and nonlinear autocorrelations that are present therein. We contend however that the use of complimentary methods is a highly recommended mean to prevent misleading or speculative interpretations.

Acknowledgements

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Appendix A. Signals of (non)linear stochastic processes

Fig. A8 depicts examplary time series of all the simulated signals generated in Section 2.1.1.

Appendix B. Partial least squares statistical analysis on EEG data and their surrogates

B.1. Methods

Partial least squares (PLS) analysis is a powerful multivariate statistical tool that aims at revealing the relationship between two blocks of datasets, in the present case age groups (young and old adults) and structural variability measures (MSE and MRMSSD). The method is based on the decomposition of the covariance of these two blocks in a set of new variables, namely latent variables (LVs), that optimally – in a least square sense – relate them, and explain as much of covariance with as few dimensions as possible. Here, we give a brief description of this technique, and refer the reader to McIntosh et al. (1996) and McIntosh and Lobaugh (2004).

In this study, we used the mean-centering Task PLS (mc-Task PLS) approach, which allows us to explore the LV's contrast of the experimental design that optimally covaries with our metrics. The method starts by constructing a brain data matrix for each experimental group. Condition averages are taken across conditions (i.e., original and surrogate data), group matrices are concatenated and the grand averaged is removed (all processing is performed in parallel for each data element). The resulting mean-centered matrix undergoes a singular value decomposition yielding three new matrices: (i) the task LVs, that describe the relations among the conditions and groups of our design, (ii) the brain LVs, that are proportional to the covariance of each data element, and (iii) singular values, that are proportional to the covariance explained by each LV's contrast. In particular, the ratio of each squared singular value by the sum of squares of all singular values of all LVs, describes the percentage of covariance that each LV accounts for. The number of resulting singular values, one for each contrast, depends on the degrees of freedom of the design. In our design we had two groups, namely young adults (YA) and old adults (OA), and two conditions, i.e., 'original' and 'surrogate' data. Thus, mc-Task PLS returned, number of conditions $\times$ number of groups $- 1 = 2 \times 2 - 1 = 3$ orthonormal LV's contrasts. Furthermore, we computed brain scores that indicate the strength of the task effect of each contrast per participant and condition. In other words, the brain score, of a particular participant for a specific contrast and condition, is the covariation of the brain data of this participant for that condition with the corresponding brain LV vector of the contrast in question.

PLS addresses the problem of multiple comparisons for statistical significance via a permutation test and, the problem of element-wise reliability via a bootstrap resampling test. The permutation test is performed on the singular values with resampling of the initial data matrices across conditions and groups, without replacement. This permutation test yields a p-value for each task LV, i.e., for each contrast. We used an alpha value of significance equal to 0.05 throughout this analysis. For the bootstrap test, the initial data matrix is resampled with replacement within conditions and groups. For the contrasts we present here, we plotted the task LVs, together with the mean brain scores and their intervals of 95% confidence that are derived from the bootstrap

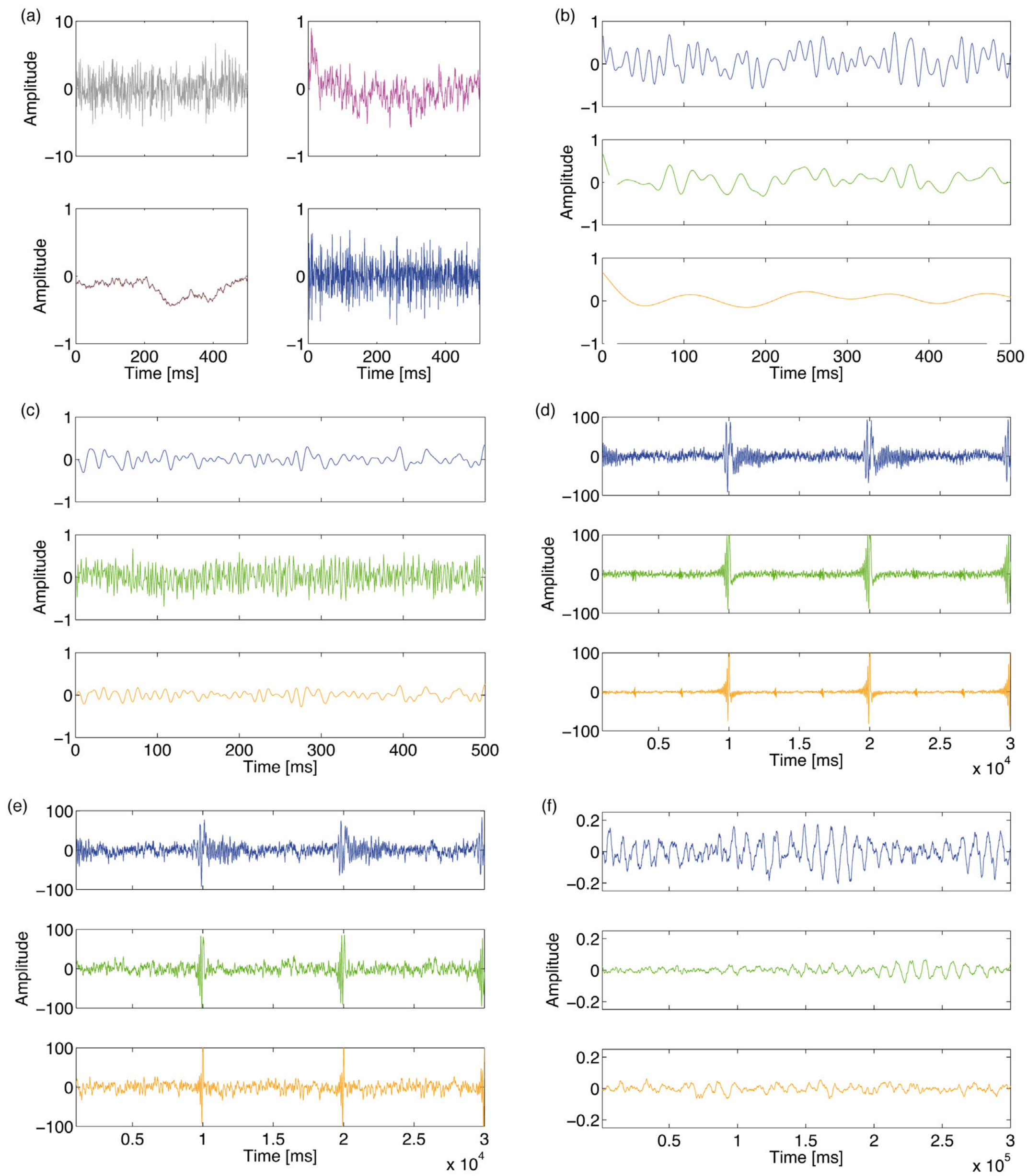

Fig. A8. Time series of the processes used in this study: (a) colored noise, (b) and (c) filtered white noise, (d) and (e) deterministic and stochastic superimposed van der Pol oscillators in limit cycle state, respectively, and (f) stochastic superimposed van der Pol oscillators in stable state. Color conventions are the same as in Figs. 1, 3–5 and 6, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

test, after mean-centering and normalizing with the respective singular value. Conditions or groups, the confidence intervals of which do not overlap, are reliably distinguished by a contrast (see Figs. B.9a, b and B.10aa). For the brain LVs, we calculated bootstrap ratios by dividing each element with its standard error as computed by the corresponding bootstrap sample distribution. Bootstrap ratios greater than 2.5758 (in absolute value) approximate the 99th two-tailed percentile for a particular element (see Figs. B.9c, d and B.10b).

B.2. Results

The PLS analysis for MSE (Fig. B9) revealed the existence of two significant LVs ($p < 0.005$ and $p < 0.003$, respectively) explaining together almost all covariance of the data (88.9% and 10.5%, respectively). The first, and dominant LV, describes a reliable main age group effect, only weakly modulated by the conditions; whereas, the second LV reliably separates the two conditions, together with a weak interaction with age groups.

We should note that, the contrasts do not depict accurate main effects, due to the explorative character of mean-centered PLS, which is free to rotate to maximize the covariance explained.

To evaluate the statistically reliable effects, as well as the statistically un- or less-reliable tendencies, we here present the brain LVs (Fig. B9c and d). With respect to the first LV, relative to young adults, MSE is higher for the older participants at time scales shorter than 24 ms, and lower at longer scales from this point on. This effect is stronger at the centrofrontal channels, and statistically reliable at least up to the scale 80 ms. For the second LV, MSE is higher for the surrogate data than original ones for fine time scales, and in particular around the global peak at 24 ms. However, above this global peak, i.e., for long time scales, the effect attenuates and slightly reverses, but it is hardly reliable for those long scales.

Unlike MSE, the PLS analysis for MRMSSD (Fig. B10) does not capture any differences between original and surrogate data. The

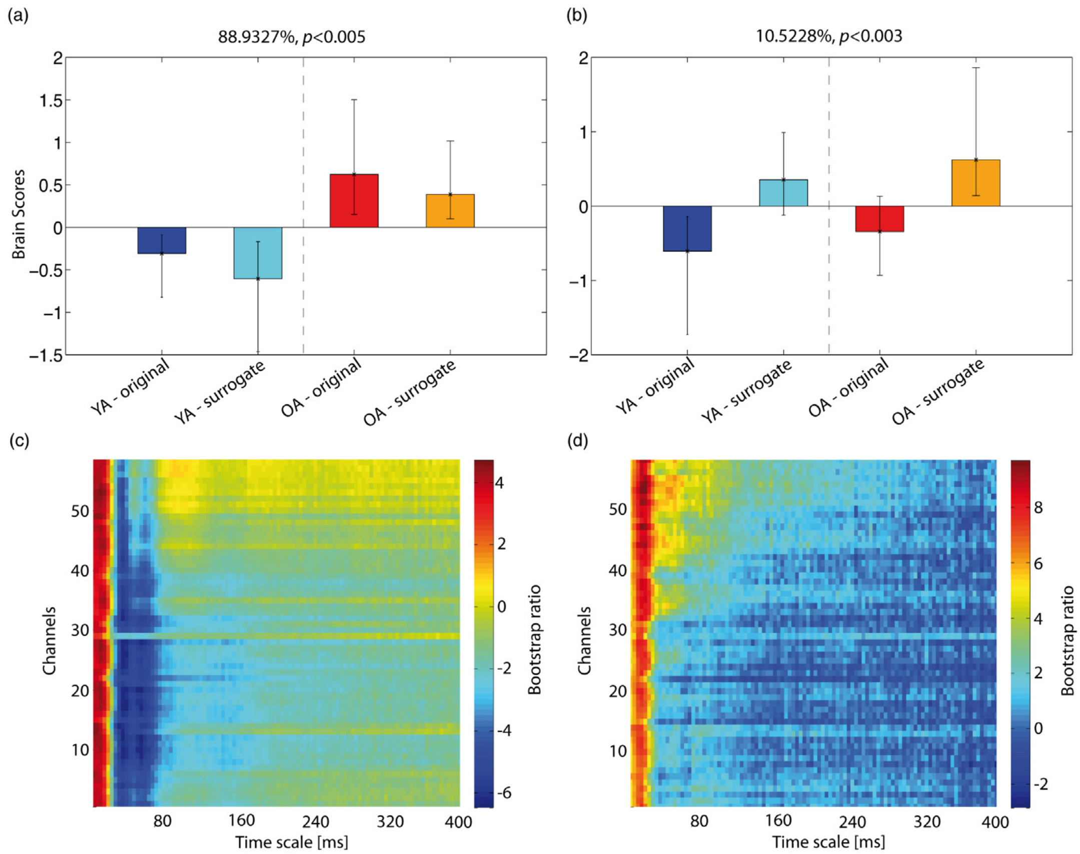

Fig. B9. Mean-centered task PLS results on MSE. The top panels (a) and (b) show the weights of the first and second task LVs of the contrast that corresponds to the group main effect Y–O and condition main effect original-surrogate, respectively. Color conventions are identical to previous Fig. 7. The percentage of covariance and the *p*-value (as derived from the parametric test for significance) are shown on top of the respective panel. Nonoverlapping confidence intervals signify that condition and/or groups are separated reliably by the contrast. The bottom panels (c) and (d) show how much each MSE's data point, covaries with the contrast that corresponds to the group/condition main effect, in terms of bootstrap ratios. Absolute values > 2.5758 approximate 99th two-tailed percentile. The vertical axis, for both panels, depicts channels arranged from top to bottom, starting from frontal and left hemisphere channels, to occipital and right hemisphere ones. The horizontal axis depicts time scale in linear scale. Reddish colors correspond to condition and/or group with positive weights of the task LVs, and the inverse for bluish colors. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

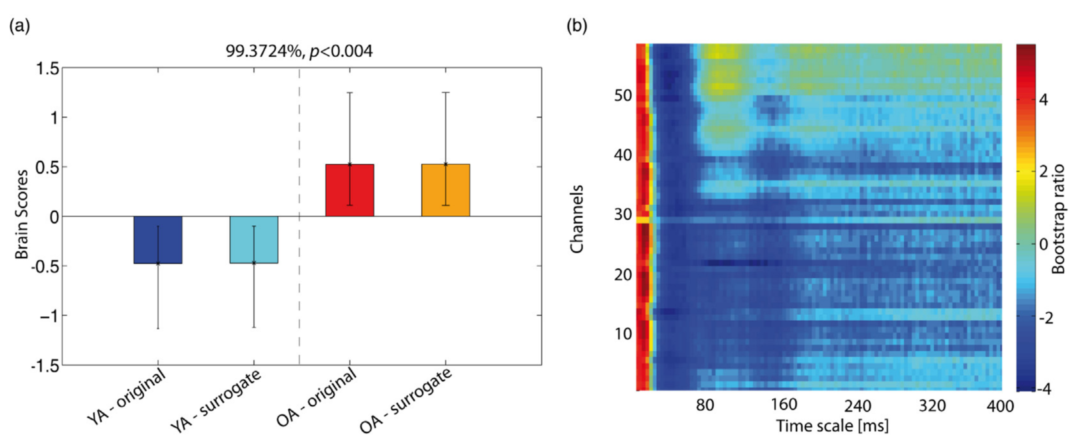

Fig. B10. Mean-centered task PLS results on MRMSSD. (a) Task latent variable, and (b) brain latent variable for the group main effect. This figure has the same arrangement and conventions as in Fig. B9. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article).

only significant LV ($p < 0.004$), that explains almost all covariance of the data (99.4%), corresponds to a main effect of age group, in which old adults are reliably separated from young ones. The pattern of brain differences, shown in the corresponding bootstrap ratios of the brain LV (Fig. B10b), closely follows the related results of MSE, only being stronger for longer time scales (for which young adults have higher MRMSSD than old ones) and showing almost no modulation across channels.

B.3. Conclusion

The PLS analysis showed that, there are statistically significant MSE differences between original and surrogate data, which point towards the existence of nonlinearities in the EEG data. However, the LV in question (i.e., second LV) explains little variance in the data, indicating that, these nonlinearities are rather weak. Moreover, the dominant latent variable is weakly modulated by any original versus surrogate data differences, which probably means that the age group effect hardly interacts with nonlinearities. At the same time, MRMSSD reproduces the age group effect equally or even stronger than MSE, and without any modulation by the original versus surrogate data differences, as expected from a metric that can reliably evaluate only linear autocorrelations.

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